

Geometry Lead (Developer 1)

CAM/AUTO | Mumbai | Full time | Job description, 29 September 2026

Company	CAM/AUTO, the in-house CAM software of a precision CNC machine shop
Location	Mumbai. On site preferred; remote considered for a strong candidate, with monthly shop visits
Type	Full time. Six-month go-live plan with a completion bonus at the week-21 gate; ongoing role in phase two
Reports to	Founder (product owner). Works daily with the process owner (senior CNC programmer) and the platform engineer
Compensation	INR 40 to 50 lakh a year (Mumbai 2026 band, to be confirmed), plus a completion bonus tied to the week-21 first-article gate

The role in one paragraph

You will replace the geometry engine of a working CAM application without rewriting the application. Today the app reads a STEP file, a drawing and a tool library and produces verified NC programs for 3-axis milling, but it computes on a 2.5D height map and recognises features from a 0.4 mm raster, which caps its accuracy at about 0.3 mm. You will make it exact: recognise features from the B-rep, offset contours with true arcs, and add an independent re-simulation of the posted NC. Your success is measured on a machine, not in a demo: 27 of 30 real parts to drawing tolerance on the first article, zero gouges.

What you will do

Weeks 1 to 3: take ownership

- Read the existing engine (12,600 lines of JavaScript, now split into modules) and its 14 test suites; write the design note that says what you will keep, wrap and replace.
- Agree the week-11 gate with the founder and the process owner: exact paths verified to 0.01 mm on 15 parts.

Weeks 3 to 11: exact geometry

- Expose the full OpenCascade API (WebAssembly build) in the app and build feature recognition directly on the B-rep: holes (through, blind, counterbored, tapped, fits), planes, outlines, pockets, islands, bosses, fillets, chamfers. Every feature carries its exact axis, radius, depth and the STEP face IDs it came from.
- Replace raster-traced contours with exact 2D offsetting with arcs (Clipper2 or equivalent) on every wall, extending what release R78 already did for the outer profile to pockets, islands and fillets.
- Define tolerances and the numerical tests: every recognised feature and every path is checked against the CAD file itself, never against the app's own output.

Weeks 8 to 15: independent check

- Design and build the second check: a parser for the posted G-code and an independent material-removal simulation (dexcel or equivalent) with tool holder and fixture, that flags gouges and collisions the app's own simulation cannot see.
- Validate it against a known-bad program and against dry runs on the shop's machines.

Weeks 12 to 26: shop loop and go-live

- Own every geometry defect found on first articles until root cause; each CMM deviation becomes a test.
- Provide the geometry side of the learning module: the offset-compensation interface that learned size corrections plug into, with clamps the process owner sets.
- Review every merge that touches geometry, including all Claude Code output.

Technical skills and technologies

Core: must have, tested in the paid task

Area	Technologies	Level	Used for
B-rep CAD kernel	OpenCascade (OCCT 7.x): TopoDS, BRep_Tool, GeomAdaptor, BRepAdaptor, ShapeAnalysis, STEPControl; or Parasolid / ACIS / CGM with willingness to move to OCCT	Expert	Feature recognition from the exact STEP model: holes, planes, pockets, fillets, chamfers
2D computational geometry	Polygon offsetting with arcs (Clipper2, CGAL, or own implementation), Boolean operations, arc fitting, tangent continuity, self-intersection removal	Expert	Exact toolpaths on every wall to 0.01 mm
Numerical methods	Floating-point tolerance, robust predicates, least-squares fitting of circles and lines, kd-trees / spatial hashing	Strong	Verifying paths against the CAD file, snapping, tests
TypeScript / JavaScript	ES2020+, TypeScript strict mode, ES modules, esbuild, Node 20	Strong	The app is TypeScript; the engine lives in it
C++	C++17, CMake, Emscripten / WebAssembly builds	Working	Building and extending the OCCT WASM module; reading kernel source

Area	Technologies	Level	Used for
Testing	Node test runner or Vitest, golden-file tests, property-based tests for geometry, Playwright for browser end-to-end	Strong	Every feature and path proven against an independent truth
Version control and CI	Git, GitHub pull-request review, GitHub Actions with a self-hosted runner	Working	Nothing merges without CI green and a human review

CAM and manufacturing: strongly preferred

Area	Technologies	Level
Toolpath generation	2.5D pocketing and contouring, helical interpolation, drill and tap cycles, rest machining, adaptive / trochoidal roughing, lead-in and lead-out, cutter compensation (G41/G42, wear offsets)	Strong
NC programming	Fanuc / Haas ISO G-code, Siemens Sinumerik (840D); can read a program line by line and predict the cut	Working
Material-removal simulation	Dexel or voxel models, swept-volume checks, gouge and collision detection with holder and fixture	Working or better
Reference tools	Has used a commercial CAM seat (Fusion 360, Mastercam, NX CAM, hyperMILL, SolidCAM or similar) and a controller simulator (CIMCO, Vericut, NCSIMUL)	Working
Metrology	Reads a CMM report and a drawing with GD&T; understands H7 fits, positional tolerance and datums	Working

Platform: working knowledge, the platform engineer leads

Area	Technologies
Rendering	three.js, WebGL basics
Documents	pdf.js (drawing text layer)
Data	PostgreSQL or SQLite, JSON schemas, REST over Fastify (Node) or FastAPI (Python)
Python	Python 3.11, pythonocc / OCP as the OCCT fallback service, pandas for reading measurement data
Containers	Docker Compose on an on-site Linux server
AI-assisted development	Claude Code: writes tasks with acceptance tests, reviews generated diffs, catches numerical errors

Not required

Machine learning frameworks (the learning module is scikit-learn on tabular data and belongs to developer 3); front-end frameworks (React, Vue: the UI is plain DOM and stays); cloud platforms (everything runs on site).

Experience profile

- 6 to 10 years, of which at least 3 in CAD kernel, CAM, CNC simulation, robotics path planning or a comparable geometry-heavy domain.
- Shipped geometry code that a shop, a vendor product or an open-source community relied on. Contributions to OCCT, CadQuery, OpenCAMLlib, Clipper, FreeCAD Path or LinuxCNC count.
- Likely backgrounds: CAM/CAD vendor teams in India (Autodesk, Siemens DISW, Dassault, PTC, Hexagon, Geometric/HCL), CAM start-ups, IIT computational geometry groups, robotics motion-planning teams.
- Degree in mechanical engineering, computer science or mathematics; a master's or PhD in computational geometry is a plus, not a requirement.
- Has stood next to a CNC machine while their program ran, and been wrong at least once.

How we hire

1. 30-minute call with the founder: the plan, the gate, your questions.
2. **Paid one-day task** on real repository material: produce the exact tool-centre offset of the D-plate silhouette (lines and arcs) to 0.01 mm, with tests that prove it against the source geometry. We pay for the day whether or not we continue.
3. Two-hour review of your task with the founder and the process owner, at the shop, in front of the machine.
4. Reference calls. Offer within a week of step 3.

What you get

- A real product with 79 releases of shop findings, a regression corpus and a shop floor that will cut your paths within days.
- A small team, a clear gate, and no committee: you own geometry.
- Claude Code on your desk, an on-site workstation (16 cores, 64 GB), machine and CMM time booked weekly.
- Completion bonus at the week-21 gate and a phase-two role leading freeform, 4-axis and lathe on the new kernel.

Keywords: OpenCascade, OCCT, B-rep, STEP, feature recognition, computational geometry, polygon offsetting, Clipper2, arc fitting, toolpath, 2.5D CAM, G-code, Fanuc, Haas, Siemens Sinumerik, cutter compensation, dexel simulation, gouge detection, TypeScript, WebAssembly, Emscripten, C++17, Node.js, Playwright, GitHub Actions, three.js, CNC milling, CMM, GD&T.